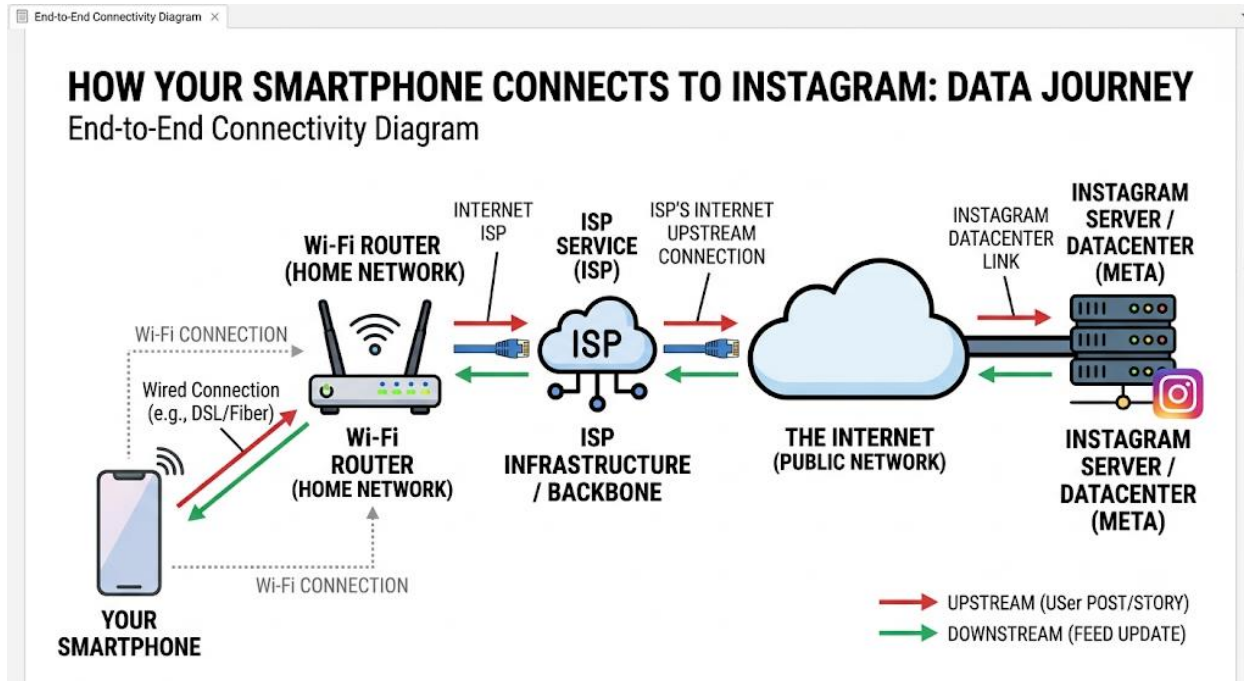


ASSIGNMENT : 9 – NETWORKING TECHNOLOGIES

GUIDED BY : VIPUL MISTRI

Q.1 Draw a simple network diagram showing how your smartphone connects to Instagram — include your device, Wi-Fi router, ISP, and Instagram's server, and label each part.

ANS :



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Q.2 List the IP address of your device and your default gateway using the 'ipconfig' command (Windows) or 'ifconfig' (Linux/Mac), and explain what each means in 1-2 lines.

ANS :

Here is how to view your IP address and default gateway on Windows using the ipconfig command, along with what each address means:

Step 1: How to find them

1. Press Win + R, type cmd, and press **Enter**.
2. Type ipconfig and press **Enter**.
3. Look for your active network adapter (e.g., *Wireless LAN adapter Wi-Fi* or *Ethernet adapter*).

Plaintext

Wireless LAN adapter Wi-Fi:

IPv4 Address. : 192.168.1.50

Subnet Mask : 255.255.255.0

Default Gateway : 192.168.1.1

Step 2: Explanation of Terms

- **IPv4 Address (e.g., 192.168.1.50):** The unique logical identifier assigned to your specific Windows device on your local network (LAN) so other local devices can send data directly to it.
- **Default Gateway (e.g., 192.168.1.1):** The IP address of your home Wi-Fi router, which acts as the exit door for all network traffic destined outside your local network (such as accessing Instagram or internet web servers).

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Q.3 Simulate a Zomato food order journey by writing the steps/data packets that travel from your device to Zomato's server and back, mentioning protocols like HTTP, TCP/IP, and DNS wherever relevant. Hint: Think about what happens when you search for a restaurant and place an order.

ANS : Here is a simplified summary of the Zomato food order data journey:

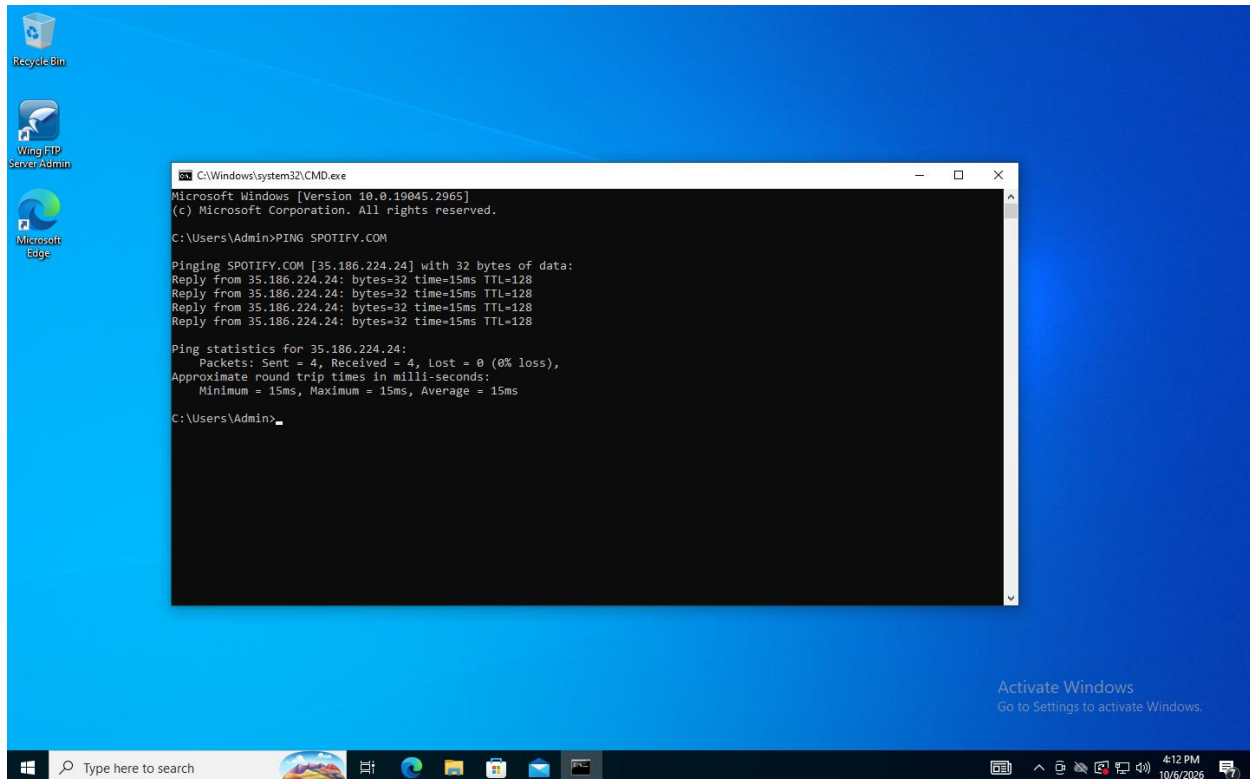
1. **DNS Lookup (UDP):** Your phone sends a DNS request to ask, *"What is the IP address for zomato.com?"* The DNS server responds with Zomato's server IP address.
2. **Connection Setup (TCP/IP):** Your device performs a TCP 3-way handshake (SYN, SYN-ACK, ACK) to establish a reliable, secure connection with Zomato's server over port 443 (HTTPS).
3. **Restaurant Search (HTTP GET):** When you search for food, your app sends an HTTP GET request. The server processes it and sends back a JSON response listing local restaurants.
4. **Placing the Order (HTTP POST):** When you click "Place Order", your phone sends an HTTP POST request containing your order details, address, and payment info. The server confirms the order and saves it to the database.
5. **Live Tracking (WebSockets / Push Notifications):** Once confirmed, a persistent WebSocket connection streams live delivery updates and driver GPS coordinates straight to your screen.

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Q.4 Use the 'ping' command to check the connection between your computer and www.spotify.com, then explain what the results mean for network troubleshooting.

ANS : Here is how to run the ping command and use its output to troubleshoot network connectivity:



Step 1: Run the Command

1. Open your terminal:
 - **Windows:** Press Win + R, type cmd, and press **Enter**.
 - **Linux/macOS:** Press Ctrl + Alt + T (or open Terminal).
2. Type the following command and press **Enter**:
 - **Windows:** ping www.spotify.com
 - **Linux/macOS:** ping -c 4 www.spotify.com (*limits output to 4 requests*)

Step 2: Example Output

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Plaintext

Pinging e9220.dscg.akamaiedge.net [104.18.22.45] with 32 bytes of data:

Reply from 104.18.22.45: bytes=32 time=18ms TTL=57

Reply from 104.18.22.45: bytes=32 time=19ms TTL=57

Reply from 104.18.22.45: bytes=32 time=17ms TTL=57

Reply from 104.18.22.45: bytes=32 time=18ms TTL=57

Ping statistics for 104.18.22.45:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 17ms, Maximum = 19ms, Average = 18ms

Step 3: What the Results Mean for Troubleshooting

- **DNS Resolution Test:** If ping www.spotify.com resolves the domain name into an IP address (e.g., 104.18.22.45), your **DNS lookup is working correctly**. If it returns *"Ping request could not find host..."*, you have a DNS resolution issue.
- **Network Reachability:** Getting Reply from... confirms basic **Layer 3 (IP) connectivity** between your host and Spotify's server (or CDN edge server).
- **Latency (time=18ms):** Indicates the round-trip time (RTT) for data packets. Low times (below 50 ms) mean a fast, responsive connection; high times (200+ ms) indicate network congestion or high physical distance
- **Packet Loss (0% loss):** Shows whether packets are dropping along the path. Any lost packets (e.g., 25% or 50% loss) suggest faulty hardware, bad Wi-Fi signals, or network saturation.
- **Time to Live (TTL=57):** An inverted hop-counter used to prevent infinite routing loops. Each router along the path decrements this value by 1 before forwarding the packet.